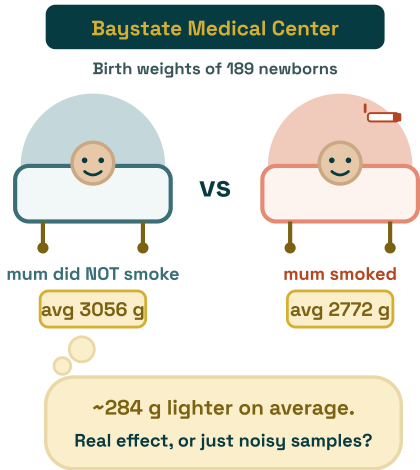


# Intro to the two-sample t-test



### The Data

🏥 189 births at one US hospital.

⚖️ Weigh each newborn.

👤 Split by whether mum smoked.

### Group means

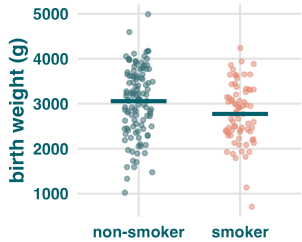
```
library(dplyr)
data(birthwt, package = "MASS")

birthwt <- birthwt |>
  mutate(smoke = factor(smoke,
    labels = c("non-smoker", "smoker")))

birthwt |>
  group_by(smoke) |>
  summarise(mean_bwt = mean(bwt))

#> smoke      mean_bwt
#> non-smoker    3056
#> smoker       2772
```

```
ggplot(birthwt, aes(smoke, bwt)) +
  geom_jitter(width = .14, alpha = .5) +
  stat_summary(fun = mean, geom = "crossbar")
```



Two independent groups. Teal bars: group means (3056 g vs 2772 g).

Smokers’ babies look ~284 g lighter. Is that real, or just two noisy samples?

### Hypothesis test framework

A two-sample test compares **two group means**.  $H_0$  says they are equal.  $H_0$  is the status quo.  $H_1$  is what the data would need to give evidence for.

#### NULL HYPOTHESIS

$H_0 : \mu_{\text{non}} = \mu_{\text{smoke}}$   
the true group means are equal.

#### ALTERNATIVE HYPOTHESIS

$H_1 : \mu_{\text{non}} \neq \mu_{\text{smoke}}$   
the true group means differ.

### Quick check: name the hypotheses

Comparing mean cholesterol in two separate groups (drug vs placebo), what is  $H_0$ ? **✓ B**

A.  $\mu_{\text{drug}} < \mu_{\text{placebo}}$   
B.  $\mu_{\text{drug}} = \mu_{\text{placebo}}$   
C.  $\mu_{\text{drug}} \neq \mu_{\text{placebo}}$

You want to detect any birth-weight difference, up or down. Which  $H_1$ ? **✓ C**

A.  $\mu_{\text{non}} > \mu_{\text{smoke}}$   
B.  $\mu_{\text{non}} < \mu_{\text{smoke}}$   
C.  $\mu_{\text{non}} \neq \mu_{\text{smoke}}$

### Measuring the gap: is 284 g a lot?

By itself, **284 g is unclear**. We weigh it against how much birth weights vary **within** each group, and how many babies are in **each** group.

#### Spread within groups

Wide spread: gap unclear

Tight spread: gap clear

#### Two sample sizes

few babies: gap may be luck

many babies: gap is trustworthy

For the same gap between means, the evidence is stronger when the within-group spread is smaller (smaller / larger) and the group sizes are larger (larger / smaller). **✓ smaller • larger**

### Step 1: spread and sample sizes

Mean gap already found: **283.8 g**. For Welch’s test, collect each group’s SD and  $n$ .

```
s1 <- sd(filter(birthwt, smoke == "non-smoker")$bwt)
n1 <- sum(birthwt$smoke == "non-smoker")
s2 <- sd(filter(birthwt, smoke == "smoker")$bwt)
n2 <- sum(birthwt$smoke == "smoker")
#> s1 = 752.7, n1 = 115
#> s2 = 659.6, n2 = 74
```

### Step 2: standard error of the difference

Combine each group’s wobble. Welch uses separate variances, not a pooled variance.

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$
$$= \sqrt{\frac{752.7^2}{115} + \frac{659.6^2}{74}}$$
$$= 104 \text{ g}$$

```
se <- sqrt(s1^2/n1 + s2^2/n2)
#> 104.0
```

### Step 3: $t$ statistic

Standardised gap: how many SEs apart are the means?

$$t = \frac{\bar{x}_1 - \bar{x}_2}{SE} = \frac{283.8}{104} = 2.73$$

```
t <- 283.8 / se
#> t = 2.73
```

### Step 4: $p$ -value

Compare  $t$  to a  $t$  curve,  $df \approx 170$  (Welch). Two-sided  $H_1$ , so  $p$  = **both tails**.

```
2 * pt(-abs(t), df = 170)
#> 0.007
```

$t = 2.73$ , two-sided  $p = 0.007$ .

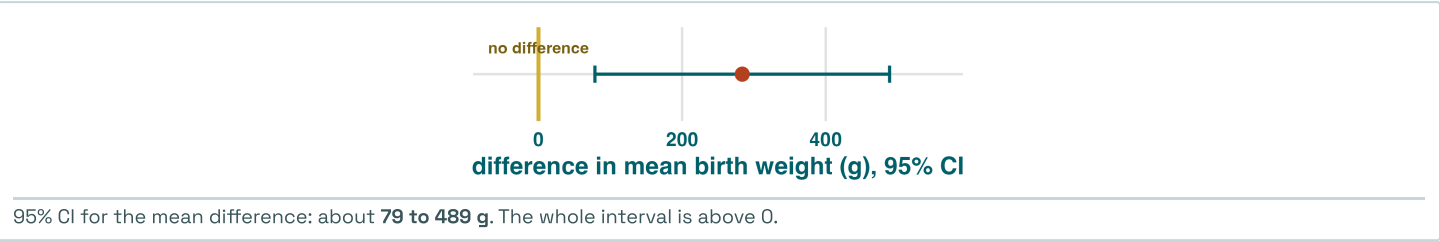
### All-in-one in R

```
t.test(bwt ~ smoke, data = birthwt); t.test(birthwt$bwt ~ birthwt$smoke) # same
# Welch Two Sample t-test
data: bwt by smoke
t = 2.7299, df = 170.1, p-value = 0.007003
alternative hypothesis: true difference in means between group non-smoker and group smoker is not equal to 0
95 percent confidence interval:
 78.57486 488.97860
sample estimates:
mean in group non-smoker    mean in group smoker
      3055.696              2771.919
#> p < 0.05 -> reject H0
```

# Confidence intervals, p-values, and reading a two-sample t-test

## Confidence interval for the birth-weight gap

The estimate was non-smoker mean minus smoker mean. Positive values mean the non-smoker group had heavier babies on average.



|               | p-value         | 95% confidence interval             |
|---------------|-----------------|-------------------------------------|
| Question      | Is 0 plausible? | Which values are plausible?         |
| Reject at .05 | $p < .05$       | CI excludes 0                       |
| Report        | $p = .007$      | mean diff = 284 g, 95% CI [79, 489] |

1. In a small clinic study, patients in a walking programme had mean resting heart rates 4 beats per minute lower than controls, with a 95% CI of [1, 7]. What decision matches at  $\alpha = 0.05$ ?

A. We fail to reject  $H_0$ .      B. We reject  $H_0$ .      C. We have proven  $H_0$  is false.

2. Reading two CI pictures: which interval gives stronger evidence of a difference?

A. The interval that crosses 0 gives stronger evidence.      B. The interval entirely above 0 gives stronger evidence.      C. The two intervals give identical evidence.

3. Reading a paper: mean systolic blood pressure was 6 mmHg lower on a new drug, 95% CI [-1, 13],  $p = .08$ . What is the clean reading?

A. The study gives clear evidence of a drug effect.      B. The result is still compatible with no true mean difference.      C. The study proves the drug has no effect.

4. Writing a manuscript: "mean difference = 284 g" without a CI is missing what?

A. The sentence is missing precision around the estimate.      B. The sentence is missing the group labels only.      C. The sentence is missing the sample mean.

- Answers 1-4**
- B. The confidence interval excludes 0, so we reject  $H_0$  at .05.
  - B. An interval entirely above 0 means every plausible value is a positive difference.
  - B. The confidence interval includes 0 and  $p > .05$ , so the result is compatible with no true mean difference.
  - A. A confidence interval shows precision by showing how uncertain the 284 g estimate is.

## ToothGrowth data

The built-in datasets  `ToothGrowth` records tooth length in two separate supplement groups: 30 guinea pigs got orange juice (OJ), 30 got ascorbic acid (VC). This is not paired data.

```
t.test(len ~ supp, data = ToothGrowth); t.test(ToothGrowth$len ~ ToothGrowth$supp) # same
```

**Welch Two Sample t-test**  
**data:** len by supp  
**t** = 1.9153, **df** = 55.309, **p-value** = 0.06063  
**alternative hypothesis:** true difference in means is not equal to 0  
**95 percent confidence interval:**  
-0.171 7.571  
**sample estimates:**  
mean in group OJ    mean in group VC  
20.66                    16.96

vitamin C, two ways

OJ                    tooth length                    VC

1. What is the null hypothesis for comparing the two separate supplement groups?

A.  $\mu_{OJ} = \mu_{VC}$   
B.  $\mu_{OJ} \neq \mu_{VC}$

2. According to the sample estimates, what is the mean tooth length in the OJ group?

A. The OJ mean is 16.96.      B. The OJ mean is 20.66.      C. The OJ mean is 1.92.

3. Match each term to its value, then to its plain-English meaning. Some values are rounded.

| TERMS   | VALUES           | MEANINGS                      |
|---------|------------------|-------------------------------|
| t       | a. 55.3          | a. # of SEs apart             |
| df      | b. 0.061         | b. adjusted df                |
| p-value | c. [-0.17, 7.57] | c. plausible range            |
| 95% CI  | d. 1.92          | d. P(t this extreme   $H_0$ ) |

4. At  $\alpha = 0.05$ , what decision should we make for the ToothGrowth test?

A. We fail to reject  $H_0$  because  $p = 0.061$ .      B. We reject  $H_0$ .      C. We have proven  $H_0$  is true.

**Answers 1-4**

- A. The null says the two true supplement means are equal.
- B. The sample estimates show OJ mean = 20.66.
- A.  $p = 0.061 > .05$ , so fail to reject  $H_0$ .

| Term    | Value            | Meaning                       |
|---------|------------------|-------------------------------|
| t       | d. 1.92          | a. # of SEs apart             |
| df      | a. 55.3          | b. adjusted df                |
| p-value | b. 0.061         | d. P(t this extreme   $H_0$ ) |
| 95% CI  | c. [-0.17, 7.57] | c. plausible range            |